

P5 — Unsupervised Concept Discovery and Evaluation for Medical Vision–Language Models

Explainable and Trustworthy AI Course

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Project. This project investigates unsupervised concept discovery and quantitative evaluation of interpretability in medical Vision–Language Models, building upon the MedConcept framework.

Overview.

Medical Vision–Language Models (VLMs) have recently achieved strong performance across a variety of clinical tasks, including disease classification, segmentation, and report generation. However, these models rely on high-dimensional and polysemantic latent representations that remain largely opaque, limiting their trustworthiness in safety-critical applications [4]. Traditional interpretability techniques, such as gradient-based saliency maps or attention visualizations, provide localized explanations but often fail to capture the semantic structure of learned representations and are typically tied to specific tasks [7]. As a result, they offer limited insight into how models internally organize medical knowledge.

Concept-based explainability has emerged as a promising direction to address these limitations by expressing predictions in terms of human-understandable concepts [8, 9]. More recently, unsupervised approaches have been proposed to automatically discover such concepts from pretrained representations without relying on manual annotations. In particular, sparse autoencoders have shown strong potential for disentangling neural representations into interpretable and semantically meaningful features [3, 11].

Building on these ideas, the MedConcept framework introduces an unsupervised pipeline for extracting latent medical concepts from pretrained VLMs and grounding them in clinically meaningful textual semantics [6]. The framework leverages sparse autoencoders to identify neuron-level concept activations and aligns them with a curated medical vocabulary derived from ontologies such as UMLS [2]. Importantly, concept discovery is performed without supervision, making it applicable across tasks and datasets.

A major challenge in concept-based interpretability is evaluation. Prior work often relies on qualitative inspection, which is subjective and difficult to scale. To address this issue, recent approaches propose using large language models (LLMs) as external evaluators to assess semantic alignment between predicted concepts and textual evidence. In MedConcept, this idea is operationalized

through three quantitative metrics—*Aligned*, *Unaligned*, and *Uncertain*—which capture semantic agreement, contradiction, and ambiguity with respect to clinical reports [6].

Despite these advances, several open challenges remain. Automatically discovered concepts may lack robustness or clinical validity, evaluation may be biased by incomplete or selective clinical reports, and the reliance on pretrained VLMs and LLMs introduces potential sources of bias and misalignment. Moreover, current approaches typically treat concepts as independent entities, overlooking the structured relationships that naturally exist among medical concepts, such as anatomical hierarchies or causal dependencies.

Goal.

The goal of this project is to study unsupervised concept discovery in medical VLMs and critically analyze how such concepts can be used to improve interpretability. The project will explore both the extraction of latent concepts and the design of reliable evaluation strategies, with particular focus on the role of external language models in assessing semantic alignment.

Required analysis, implementation, and evaluation.

- **Literature Review.** Conduct a systematic review of interpretability methods for medical Vision–Language Models, with particular focus on concept-based explainability approaches such as Concept Bottleneck Models and TCAV [8], as well as recent advances in unsupervised concept discovery using sparse autoencoders and representation disentanglement [12, 5, 1, 10]. The review should also cover existing evaluation strategies for explanations, including LLM-based semantic evaluation protocols and their limitations [13].
- **Identification of Research Gaps.** Identify key limitations of current approaches, such as the lack of guarantees on the semantic validity and stability of discovered concepts, the dependence on external resources such as ontologies and large language models, and the reliance on incomplete or noisy textual reports for evaluation. Particular attention should be given to the absence of structured relationships among concepts and the challenges in quantitatively assessing interpretability in a scalable and reproducible way.
- **Implementation.** Design and implement a simplified unsupervised concept discovery pipeline inspired by recent frameworks. Possible directions include:
 - training a sparse autoencoder to extract latent concepts from pre-trained visual or multimodal embeddings;
 - aligning latent features with textual concepts using similarity-based matching within a shared embedding space;

- generating concept-based explanations for individual samples by identifying highly activated concepts;
 - exploring simple extensions such as filtering, clustering, or organizing discovered concepts into structured representations.
- **Evaluation.** Evaluate the proposed approach on at least one dataset using both qualitative and quantitative interpretability metrics. The evaluation should include analysis of concept quality and semantic alignment, for example by implementing or adapting metrics such as Aligned, Unaligned, and Uncertain scores, and by assessing the consistency and usefulness of the generated explanations. Additional analysis may investigate sensitivity to model design choices and characterize failure cases such as ambiguous or spurious concepts.

References

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